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About this Course

Key Components of Llamafile

Developing Portable CLI with Cosmopolitan

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Submitted

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Running Llamafile

Conclusion and Next Steps

Lesson Reflection

Lesson Summary

This lesson demonstrated using the Cosmopolitan framework to create a cross-platform phrase repetition command line tool in C. It showed compiling with the cosmocc toolchain, modular function design, parsing arguments with argc/argv, string/integer conversion, and portability.

Key Points

- Cosmopolitan enables portable command line binaries
- Custom compiler outputs binaries for any OS
- Separate phrase generation function
- Parse argc/argv to handle arguments
- strcmp for string comparisons
- atoi converts string numbers

Reflection Questions

1. How does Cosmopolitan provide portability at the binary level?
2. What is the benefit of splitting code into a separate phrase generator function?
3. How can argc and argv enable flexible CLI argument handling?
4. What string/number helper functions are useful for CLI programs?
5. Where else might you apply Cosmopolitan portability?

Challenge Exercises

1. Add support for more types of CLI arguments
2. Generate phrases from an external file
3. Create a user input loop instead of CLI
4. Extend functionality with other string functions
5. Build a different portable command line utility

Code Example

```
1 // Portable CLI phrase generator
2
3 #include <stdio.h>
4
5 // Generate phrases
6 void phrase_generator(int num, char *text) {
7     for(int i = 0; i < num; i++){
8         printf("%s ", text);
9     }
10    printf("\n"); // Newline
11 }
12
13 int main(int argc, char **argv) {
14
15     // Parse arguments from command line
16     int count = argc > 1 ? atoi(argv[1]) : 1;
17     char *phrase = "Hello world";
18
19     // Repeat phrase
20     phrase_generator(count, phrase);
21
22     return 0;
23 }
```

Run

Reset

Hello world

Go to next item

✔ Completed



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